

EXPERIMENT 2

Aim

To convert the ER diagram of the Indian e-commerce platform into a relational schema and implement it in MySQL using primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE, and ON DELETE SET NULL constraints, followed by sample data insertion and referential-integrity testing.

Relational Schema

Customer, Address, Seller, Category, Product, Orders, OrderItem, Payment, and Delivery.

Source Code

```
CREATE DATABASE IF NOT EXISTS EcommerceDB;
USE EcommerceDB;

CREATE TABLE Customer (
    Customer_ID INT PRIMARY KEY,
    First_Name VARCHAR(30) NOT NULL,
    Last_Name VARCHAR(30),
    Email VARCHAR(100) NOT NULL UNIQUE,
    Mobile_No VARCHAR(15) UNIQUE
);

CREATE TABLE Address (
    Address_ID INT PRIMARY KEY,
    Customer_ID INT,
    House_No VARCHAR(20),
    Street VARCHAR(50),
    Area VARCHAR(50),
    City VARCHAR(30) NOT NULL,
    State VARCHAR(30) NOT NULL,
    Pincode VARCHAR(10) NOT NULL,
    Address_Type VARCHAR(20),
    FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID)
    ON DELETE CASCADE
);

CREATE TABLE Seller (
    Seller_ID INT PRIMARY KEY,
    Seller_Name VARCHAR(50) NOT NULL,
    Business_Name VARCHAR(80),
    Email VARCHAR(100) UNIQUE,
    Phone VARCHAR(15),
    GSTIN VARCHAR(20) UNIQUE
);

CREATE TABLE Category (
    Category_ID INT PRIMARY KEY,
    Category_Name VARCHAR(50) NOT NULL UNIQUE,
    Description VARCHAR(150)
);

CREATE TABLE Product (
    Product_ID INT PRIMARY KEY,
    Product_Name VARCHAR(80) NOT NULL,
    Description VARCHAR(150),
    Price DECIMAL(10,2) NOT NULL,
    Brand VARCHAR(50),
    Stock_Quantity INT DEFAULT 0,
    Category_ID INT NULL,
    FOREIGN KEY (Category_ID) REFERENCES Category(Category_ID)
    ON DELETE SET NULL
);

CREATE TABLE Orders (
    Order_ID INT PRIMARY KEY,
    Customer_ID INT NOT NULL,
    Order_Date DATE NOT NULL,
    Order_Status VARCHAR(30),
    Total_Amount DECIMAL(10,2),
    FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID)
    ON DELETE CASCADE
);

CREATE TABLE OrderItem (
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    Order_ID INT,
    Product_ID INT,
    Quantity INT NOT NULL,
    Unit_Price DECIMAL(10,2) NOT NULL,
    Discount DECIMAL(5,2) DEFAULT 0,
    PRIMARY KEY (Order_ID, Product_ID),
    FOREIGN KEY (Order_ID) REFERENCES Orders(Order_ID) ON DELETE CASCADE,
    FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID) ON DELETE CASCADE
);

CREATE TABLE Payment (
    Payment_ID INT PRIMARY KEY,
    Order_ID INT NOT NULL UNIQUE,
    Payment_Date DATE,
    Amount DECIMAL(10,2),
    Payment_Method VARCHAR(30),
    Payment_Status VARCHAR(30),
    Transaction_ID VARCHAR(60),
    FOREIGN KEY (Order_ID) REFERENCES Orders(Order_ID) ON DELETE CASCADE
);

CREATE TABLE Delivery (
    Delivery_ID INT PRIMARY KEY,
    Order_ID INT NOT NULL UNIQUE,
    Delivery_Date DATE,
    Delivery_Status VARCHAR(30),
    Tracking_Number VARCHAR(50) UNIQUE,
    Courier_Name VARCHAR(50),
    Expected_Date DATE,
    FOREIGN KEY (Order_ID) REFERENCES Orders(Order_ID) ON DELETE CASCADE
);

INSERT INTO Customer VALUES
(101, 'Yuvraj', 'Singh', 'yuvraj@example.com', '9876543210'),
(102, 'Aman', 'Sharma', 'aman@example.com', '9876543211'),
(103, 'Riya', 'Verma', 'riya@example.com', '9876543212');

INSERT INTO Address VALUES
(1, 101, '12A', 'MG Road', 'Hazratganj', 'Lucknow', 'Uttar Pradesh', '226001', 'Home'),
(2, 102, '45B', 'Park Street', 'Central Delhi', 'Delhi', 'Delhi', '110001', 'Home');

INSERT INTO Seller VALUES
(201, 'Tech Store', 'Tech India Pvt Ltd', 'tech@example.com', '9876500001', '09ABCDE1234F1Z5'),
(202, 'Fashion Hub', 'Fashion Hub India', 'fashion@example.com', '9876500002', '09ABCDE5678F1Z6');

INSERT INTO Category VALUES
(301, 'Electronics', 'Electronic products'),
(302, 'Fashion', 'Clothing and accessories');

INSERT INTO Product VALUES
(401, 'Laptop', 'Gaming laptop', 65000.00, 'HP', 20, 301),
(402, 'T-Shirt', 'Cotton T-shirt', 799.00, 'Nike', 50, 302);

INSERT INTO Orders VALUES
(501, 101, '2026-09-01', 'Placed', 65000.00),
(502, 102, '2026-09-02', 'Delivered', 799.00);

INSERT INTO OrderItem VALUES
(501, 401, 1, 65000.00, 0.00),
(502, 402, 1, 799.00, 10.00);

INSERT INTO Payment VALUES
(601, 501, '2026-09-01', 65000.00, 'UPI', 'Successful', 'TXN1001'),
(602, 502, '2026-09-02', 719.10, 'Card', 'Successful', 'TXN1002');

INSERT INTO Delivery VALUES
(701, 501, '2026-09-05', 'Delivered', 'TRK1001', 'BlueDart', '2026-09-05'),
(702, 502, '2026-09-06', 'Delivered', 'TRK1002', 'Delhivery', '2026-09-06');

SELECT * FROM Customer;
SELECT * FROM Address;
SELECT * FROM Seller;
SELECT * FROM Category;
SELECT * FROM Product;
SELECT * FROM Orders;
SELECT * FROM OrderItem;
SELECT * FROM Payment;
SELECT * FROM Delivery;

-- Referential integrity test (expected to fail; Customer_ID 999 does not exist):

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```
-- INSERT INTO Orders VALUES (503, 999, '2026-09-10', 'Placed', 5000.00);

-- ON DELETE SET NULL test (run separately):
-- DELETE FROM Category WHERE Category_ID = 301;
-- SELECT * FROM Product;

-- ON DELETE CASCADE test (run separately):
-- DELETE FROM Customer WHERE Customer_ID = 101;
```

Expected Referential Integrity Test

The commented INSERT statement using Customer_ID 999 should fail because that customer does not exist. The ON DELETE SET NULL test sets the related Product.Category_ID to NULL. The ON DELETE CASCADE test deletes the customer's dependent Address and Orders records.

Result

The ER diagram was converted into relational tables and implemented with primary keys, foreign keys, uniqueness and nullability constraints, and referential actions.